

Tessa Adams
CariSurg MedTech Pathways Healthcare AI Program
Week 1: Final Deliverable
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I have included three project proposals as I am still deciding which direction to pursue for the 12-week pilot project. My goal is to explore how AI can be applied to specific, smaller tasks that could still meaningfully improve emergency department efficiency. I would appreciate any feedback on which proposal you believe is the most suitable and feasible within the available timeframe.

Research Methodology Note

Papers were sourced using PubMed. ChatGPT was used to help extract the Q1–Q4 structure from each abstract, discussion and conclusion. After confirming each paper was relevant to the research focus, the full text was reviewed and summaries refined. The problem statement was designed around the research gaps identified in the previous work literature, particularly the limited validation of these systems in Caribbean emergency department settings. The reference lists were manually produced using the Vancouver citation style and the example used in Week 1 Tutorial 1.

CariSurg MedTech Pathways Research Project

Preliminary Proposal 1

Project Title: AI-Assisted Early Sepsis Detection in the Emergency Department: A Pilot Study at Mercer General Hospital

Student Name: Tessa Adams

Statement of the Problem:

Sepsis is a severe reaction to infection responsible for approximately 11 million deaths worldwide each year (Rudd et al., 2020). Early recognition and treatment are critical, as studies indicate that each hour of delay in antibiotic treatment can increase mortality risk by 4–9% (AAMC, 2023). AI tools such as NAVOY® Sepsis have shown promise in detecting sepsis earlier in ICU settings in Northern Europe (Persson et al., 2024). However, these tools have not been evaluated in Caribbean emergency departments, where triage nurses rely mainly on clinical judgment and delays in sepsis recognition remain a real and ongoing challenge.

Previous Work:

Several machine learning systems have been developed for early sepsis detection, but most have been tested in North American and European hospitals rather than Caribbean emergency departments.

Summary 1 - Sepsis Prediction at Emergency Department Triage Using Natural Language Processing: Retrospective Cohort Study

Brann et al. (2024) investigated whether a machine learning model could predict sepsis at the time of emergency department triage using nursing triage notes and patient data recorded on arrival. The model was trained and tested on over one million adult emergency department encounters collected from four academically affiliated US hospitals between 2015 and 2021. The system detected 36.1% of sepsis cases three hours before the first antibiotic treatment, 49.9% two hours before, and 68.3% one hour before, indicating strong overall predictive performance (AUC = 0.94). However, the authors note that the model was developed exclusively in US academic hospitals with standardized nursing documentation, making its performance in settings such as Caribbean emergency departments uncertain.

Summary 2 - Early prediction of sepsis in intensive care patients using the machine learning algorithm NAVOY® Sepsis, a prospective randomized clinical validation study

Persson et al. (2024) investigated whether the machine learning algorithm NAVOY® Sepsis could predict sepsis in ICU patients before clinical recognition. The study evaluated 304 adult ICU patients at Skåne University Hospital in Sweden using vital signs, blood gas measurements, and laboratory data. Results showed that NAVOY® Sepsis could predict sepsis approximately three hours before clinical diagnosis with good performance (accuracy of 79%, sensitivity of 80%, and specificity of 78%). However, the authors note that the study was conducted in a single ICU and delays in manual data entry during COVID-19 affected the timing of alerts, limiting their usefulness for clinical decision-making.

Summary 3 - External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients

Wong et al. (2021) evaluated the performance of the Epic Sepsis Model, a machine learning tool widely used across US hospitals, by testing it on electronic health record data from 27,697 patients at Michigan Medicine, a different healthcare system from where the model was originally trained. The researchers assessed how accurately the model predicted sepsis, how many cases it detected, and how often it generated alerts using scores calculated every 15 minutes. Results showed poor performance, the model missed 67% of sepsis cases and generated alerts for 18% of all hospitalised patients, a false-positive rate so disruptive that clinical staff began ignoring the alerts altogether. The authors acknowledged that the study was limited to one hospital using one alert setting, so it is unclear how the model would perform in a different healthcare system, including a Caribbean emergency department like Mercer General Hospital.

Summary 4 - Global, regional, and national sepsis incidence and mortality, 1990–2017: analysis for the Global Burden of Disease Study

Rudd et al. (2020) estimated the global scale of sepsis using death records from 109 million individuals and hospital data from 8.7 million patients across 195 countries. The study found that in 2017 alone, there were approximately 48.9 million sepsis cases worldwide and 11 million sepsis-related deaths, accounting for nearly one in five deaths globally. Although the overall rates of sepsis cases and deaths decreased between 1990 and 2017, the highest rates remained in lower-income regions including sub-Saharan Africa

and south Asia. The authors noted that limited high-quality hospital and mortality data from many lower-income regions may lead to underestimation of the true global burden of sepsis.

Proposed Solution

I propose a 12-week pilot to test whether a machine learning model trained on a Mercer General ED dataset can identify at least 50% of sepsis cases at least two hours before clinical diagnosis. The model will use only information typically available at triage, such as vital signs, age, and chief complaint, to reflect the conditions of a real Caribbean ED where lab results are not always immediately available.

Gaps: Brann et al. (2024) demonstrated strong sepsis prediction at triage across four large US academic hospitals but did not test performance in settings where nursing notes are brief and digital infrastructure is inconsistent, the everyday reality at many Caribbean EDs. Persson et al. (2024) validated NAVOY® Sepsis in a well-resourced Swedish ICU but stated that generalisability to lower-resource or non-European settings remains unproven, leaving its relevance to Caribbean emergency departments untested.

References:

1. Brann F, Sterling NW, Frisch SO, Schrager JD. Sepsis prediction at emergency department triage using natural language processing: a retrospective cohort study. *JMIR AI*. 2024;3:e49784. [doi:10.2196/49784](https://doi.org/10.2196/49784)
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4. Balch B. Sepsis is the third leading cause of death in U.S. hospitals. But quick action can save lives. Association of American Medical Colleges (AAMC) News. 2023 Oct 10. <https://www.aamc.org/news/sepsis-third-leading-cause-death-us-hospitals-quick-action-can-save-lives>

Preliminary Proposal 2

Project Title: Improving Emergency Department Patient Flow Using Machine Learning Support Systems: A Pilot Study at Mercer General Hospital

Student Name: Tessa Adams

Statement of the Problem:

The Queen Elizabeth Hospital (QEH) in Barbados recently introduced bedside discharge tracking boards to improve how patients move through the hospital and how beds are managed (Graham, 2025). However, the process still relies heavily on staff communicating manually between departments, which can slow things down and contribute to overcrowding and longer wait times. Research shows that AI-assisted patient flow systems can reduce emergency department length of stay by 7.5 to 11.9 minutes per patient without increasing risk to patients (Hodgson et al., 2025). Despite these results, most systems have only been tested in large, well-staffed and well-resourced hospitals, leaving little evidence on whether they would work as effectively in a smaller Caribbean health system like QEH or Mercer General Hospital.

Previous Work:

Recent research has explored how machine learning (ML) can improve patient flow, discharge planning, and bed management. The following studies examine the benefits and limitations of these systems in hospital settings.

Summary 5- Combining machine learning and optimization for the operational patient-bed assignment problem

Schäfer et al. (2023) investigated whether machine learning (ML) could improve hospital bed assignment by predicting how many emergency patients would need admission. The system used information such as weather, dates, local events, and hospital occupancy levels to predict patient demand, then combined those predictions with an optimization algorithm to improve bed management decisions. Results showed that the ML models improved prediction accuracy by up to 17% compared to a baseline approach and improved overall hospital bed-assignment performance by up to 3.3%. However, the study was conducted using data from a single large German hospital, so its effectiveness in smaller hospitals such as Mercer General remains uncertain.

Summary 6 -Artificial Intelligence-Assisted Emergency Department Vertical Patient Flow Optimization

Hodgson et al. (2025) examined whether machine learning (ML) could improve emergency department (ED) patient flow by using a vertical processing pathway (VPP), a system where less critical patients are treated in chairs rather than beds. The researchers trained a ML model on 49,350 ED encounters to predict which patients were appropriate for vertical processing, then used those predictions to redesign patient flow during busy periods in the ED. After a 13-week trial, the hospital reduced average patient stay in the

ED by about 7.5–11.9 minutes without increasing return visits or hospital admissions. However, the authors noted several limitations including inconsistent staff adherence to the new system and the continued reliance on manual decision-making by doctors and nurses rather than full integration into the hospital's electronic health record system.

Proposed Solution

I propose a 12-week pilot study to evaluate a machine learning patient flow support system inspired by the Queen Elizabeth Hospital's bedside discharge tracking initiative. The model will be trained on de-identified Mercer patient records and use routinely collected information to flag patients who may be suitable for earlier discharge. Success will be measured by whether the model correctly identifies patients suitable for early discharge at least 75% of the time.

Gaps: Schäfer et al. (2023) showed machine learning improved bed assignment accuracy by up to 17% at a large German hospital but acknowledged the system had not been tested in smaller Caribbean hospitals. Hodgson et al. (2025) noted that inconsistent staff adherence and reliance on manual decision-making limited their system even in a well-resourced US hospital, barriers that would likely be more pronounced at a smaller Caribbean ED like Mercer General.

References

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3. El-Bouri R, Taylor T, Youssef A, Zhu T, Clifton DA. Machine learning in patient flow: a review. *Prog Biomed Eng*. 2021;3(2):022002. doi:10.1088/2516-1091/abddc5.
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Preliminary Proposal 3

Project Title: Improving Emergency Department Laboratory Communication and result turnaround Using Automated Critical Result Alerts: A Pilot Study at Mercer General Hospital

Student Name: Tessa Adams

Statement of the Problem:

Delays in laboratory results and slow communication between departments are a major cause of overcrowding and longer patient stays in emergency departments. Research shows that cutting just one minute from lab processing time reduces how long a patient spends in the ED by half a minute, and that reducing delays by 5 to 15 minutes could allow a single ED to see up to 386 more patients a year (Kaushik et al., 2018). Automated alert systems that notify doctors directly when critical test results are ready have also been shown to speed up treatment, shorten length of stay, and reduce mortality (García-de la Torre et al., 2025). In many Caribbean hospitals, where labs are often slower and staff are stretched thin, reducing these delays could significantly improve patient flow and emergency department efficiency.

Previous Work

Recent research shows that faster lab workflows and automated alert systems can improve ED efficiency, but the evidence comes almost entirely from large, well-resourced hospitals, leaving little guidance for Caribbean settings.

Summary 7 - Reduction in laboratory turnaround time decreases emergency room length of stay

Kaushik et al. (2018) investigated whether processing lab results faster could reduce how long patients spend in the emergency department, using historical patient records from several US hospitals. They found that every one-minute reduction in lab processing time reduced ED length of stay by half a minute, with consistent improvements across all patient acuity levels. They also estimated that cutting lab processing times by 5 to 15 minutes could allow a single ED to see between 127 and 386 more patients each year. However, the authors acknowledged that staffing levels, specialist availability, and how quickly doctors act on results were not accounted for, limiting how directly these findings apply to a Caribbean hospital setting.

Summary 8 - Implementation of an automated alert system of critical results in hospitalized and emergency patients

García-de la Torre et al. (2025) evaluated whether an automated alert system for critical lab results could improve patient outcomes compared to traditional telephone reporting. In a two-year case-control study, the researchers compared phone-based result notification with a secure text messaging system that sent alerts directly to physicians' smartphones. The automated system resulted in faster notification times, quicker treatment, shorter length of stay, and lower mortality rates. However, the study was conducted in

a single hospital with reliable digital infrastructure, and it is unclear how the system would perform in settings where internet stability cannot always be guaranteed, as is common in many Caribbean hospitals.

Proposed Solution

This project proposes a 12-week simulation of an automated critical laboratory result alert system using de-identified data. Using Mercer General Hospital data, the system will model how automated alerts for critical laboratory results could support faster physician notification by linking abnormal results to triage acuity levels already recorded in the data. The goal is to evaluate whether patients triaged at higher acuity levels consistently had critical results that, if flagged earlier, could have supported faster clinical decision-making. Success will be measured by how accurately the model identifies high-acuity patients with critical lab results, using outcomes already recorded in the dataset.

Gaps: García-de la Torre et al. (2025) demonstrated the benefits of automated critical result alerts in a single hospital with reliable digital infrastructure, but did not address whether the system would function in settings where strong internet connectivity cannot always be guaranteed, as is common in Caribbean hospitals. Kaushik et al. (2018) linked lab turnaround time to ED length of stay but did not account for how quickly doctors act on results once available, a factor likely just as significant as processing time in a Caribbean ED where staff are often managing multiple patients at once.

References

1. Kaushik N, Khangulov VS, O'Hara M, Arnaout R. Reduction in laboratory turnaround time decreases emergency room length of stay. Open Access Emerg Med. 2018;10:37-45. <https://doi.org/10.2147/OAEM.S155988>
2. García-de la Torre A, Sánchez-Lora FJ, Navarrete-Ortiz MC, Trillo López P, Ortiz-García C, Méndez Alarcón G, et al. Implementation of an automated alert system of critical results in hospitalized and emergency patients. Clin Chem Lab Med. 2025;64(4):838-844. <https://doi.org/10.1515/cclm-2024-0918>

General AI-assisted Triage Research Paper Summaries

Summary 9- Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review

Tyler et al. (2024) researched whether machine learning models improve clinical outcomes and triage accuracy in emergency departments compared to traditional five-level triage systems that rely on nurses' subjective judgment. Using a scoping review based on Joanna Briggs Institute methodology, the researchers searched EMBASE, Ovid MEDLINE, and Web of Science for US-based, peer-reviewed primary research studies published between 2013 and 2023, screening 1,140 citations down to 29 included studies. Overall, the findings showed that machine learning models consistently outperformed conventional triage systems in areas such as predictive accuracy, disease identification, risk assessment, resource allocation and prediction of length of stay. However, the authors noted that the review was limited to studies published in the United States, which may reduce how well the findings apply to healthcare systems with different staffing structures, patient populations, and resource availability. This represents a direct gap for Mercer General Hospital, a Caribbean emergency department with a distinct patient mix and resource constraints not considered in the 29 reviewed studies.

Summary 10- AI-driven triage in emergency departments: A review of benefits, challenges, and future directions

Da'Costa et al. (2025) explored whether artificial intelligence–driven triage systems can improve emergency department patient prioritisation compared to traditional triage methods that rely on subjective clinical judgment, particularly during periods of overcrowding and mass-casualty events. To investigate this, the authors conducted a narrative review of peer-reviewed literature published between 2014 and 2024, using databases including PubMed, Scopus, IEEE Xplore, and Google Scholar. The findings suggested that machine learning triage systems can significantly improve patient prioritisation, reduce wait times, and optimise resource allocation within emergency departments. However, the author notes that the narrative review design limits quantitative assessment of evidence strength, and may exclude relevant older, non-English, and grey literature. They also highlight the potential for publication bias and a lack of consideration for region-specific differences in healthcare systems and infrastructure. As a result, the generalisability of findings to diverse settings, including Caribbean emergency departments, remains limited.

Summary 11 - TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers

Araouchi and Adda (2024) investigated whether combining machine learning (ML) models could predict emergency department triage levels more accurately than a single model, with the aim of helping clinicians prioritise patients more effectively during periods of overcrowding. To test this, they developed and compared seven ML models (Support Vector Machine, Random Forest, Artificial Neural Network, Gradient Boosting Machine, Logistic Regression, XGBoost, and a stacking ensemble model that combined the three best-performing algorithms) using a Korean emergency department dataset labelled according to the Korean Triage and Acuity Scale (KTAS). The results showed that the combined model, TriageIntelli, achieved the highest predictive accuracy among all seven models, demonstrating that

combining the outputs of multiple well-performing algorithms improved triage classification performance compared to relying on a single model alone. However, the authors noted that the model was developed and tested only on Korean KTAS data and had not been validated in non-Korean populations. Differences in patient demographics, clinical practices, and documentation systems limit its generalisability to other healthcare settings.

References:

1. Tyler S, Olis M, Aust N, et al. Use of artificial intelligence in triage in hospital emergency departments: a scoping review. *Cureus*. 2024;16(5):e59906. <https://doi.org/10.7759/cureus.59906>
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